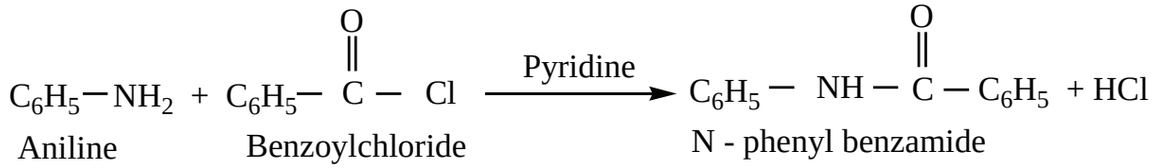
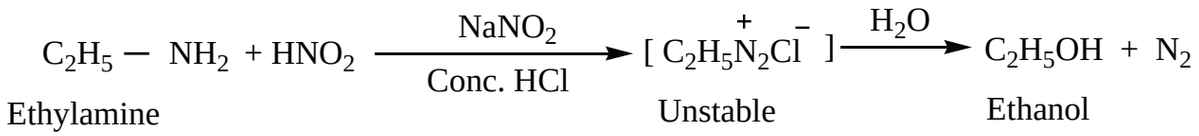


4) Reaction with nitrous acid

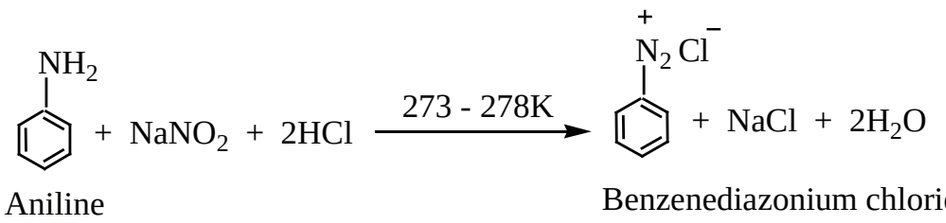
Three classes of amines react differently with nitrous acid which is prepared in situ from a mixture of $NaNO_2$ and HCl .

a) primary amines

i) Ethylamine reacts with nitrous acid to give ethyl diazonium chloride, which is unstable and it is converted to ethanol by liberating N_2.

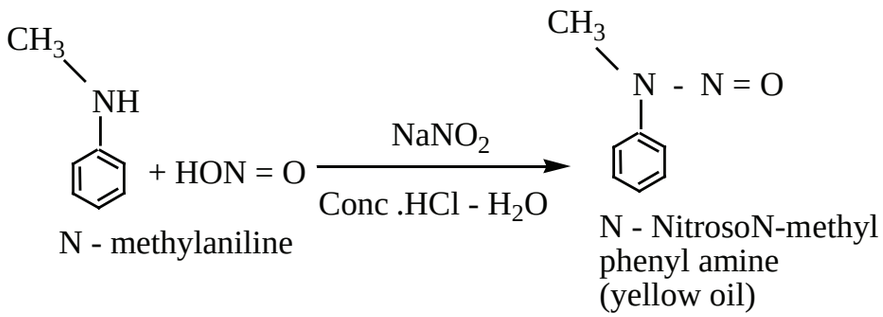


ii) Aniline reacts with nitrous acid at low temperature (273 – 278 K) to give benzene diazonium chloride which is stable for a short time and slowly decomposes even at low temperatures. This reaction is known as diazotization.



b) secondary amines

Alkyl and aryl secondary amines react with nitrous acid to give N – nitroso amine as yellow oily liquid which is insoluble in water.



This reaction is known as Libermann's nitroso test,

c) Teritiary amine

i) Aliphatic tertiary amine reacts with nitrous acid to form trialkyl ammonium nitrite salt, which is soluble in water.

XII U13-Organic Nitrogen Compounds.indd 215 11-01-2021 16:30:36